



AI-ENHANCED WASTE COLLECTION

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THE DEVELOPERS



Ezen



Felix



Katie



Henry



Marek



Alex



OUR CLIENT

We are building our project in collaboration with Gowri Shankar, co-founder of RecycleNXT.

His company is on a mission to redefine how the world thinks about waste by creating a circular economy - transforming discarded plastic bottles into long-lasting, fashionable products.



Data Collection and Reporting

Collecting data such as distance travelled, and fuel consumed to be used for the prediction model and report on environmental impact.

Prediction Model

Predict when stops need to be collected using historical data.

OUR PROJECT

iOS & Android APP

The key feature of our project is developing a cross-platform application for the drivers to access the optimised routes.

Route Planning

Plan the most efficient route between collection points to reduce the environmental impacts of waste collection

Admin Portal

A web application that allows administrators to add and update stop information.



USER STORIES



01

Waste Collectors

As a **waste collector**, I want to be able to **view the routes** easily so that I can **follow them and collect the waste**

03

Environmental Organization

As an **environmental organization**, I want to be able to **see that the routes are reducing the amount of carbon and emissions via the report**, so I can be **sure the company is helping to protect the environment**.



02

Waste Management Company/City Council

As a **city council/municipal**, I want to be able to **enter the collection points and get the most efficient route to save money and time**.

04

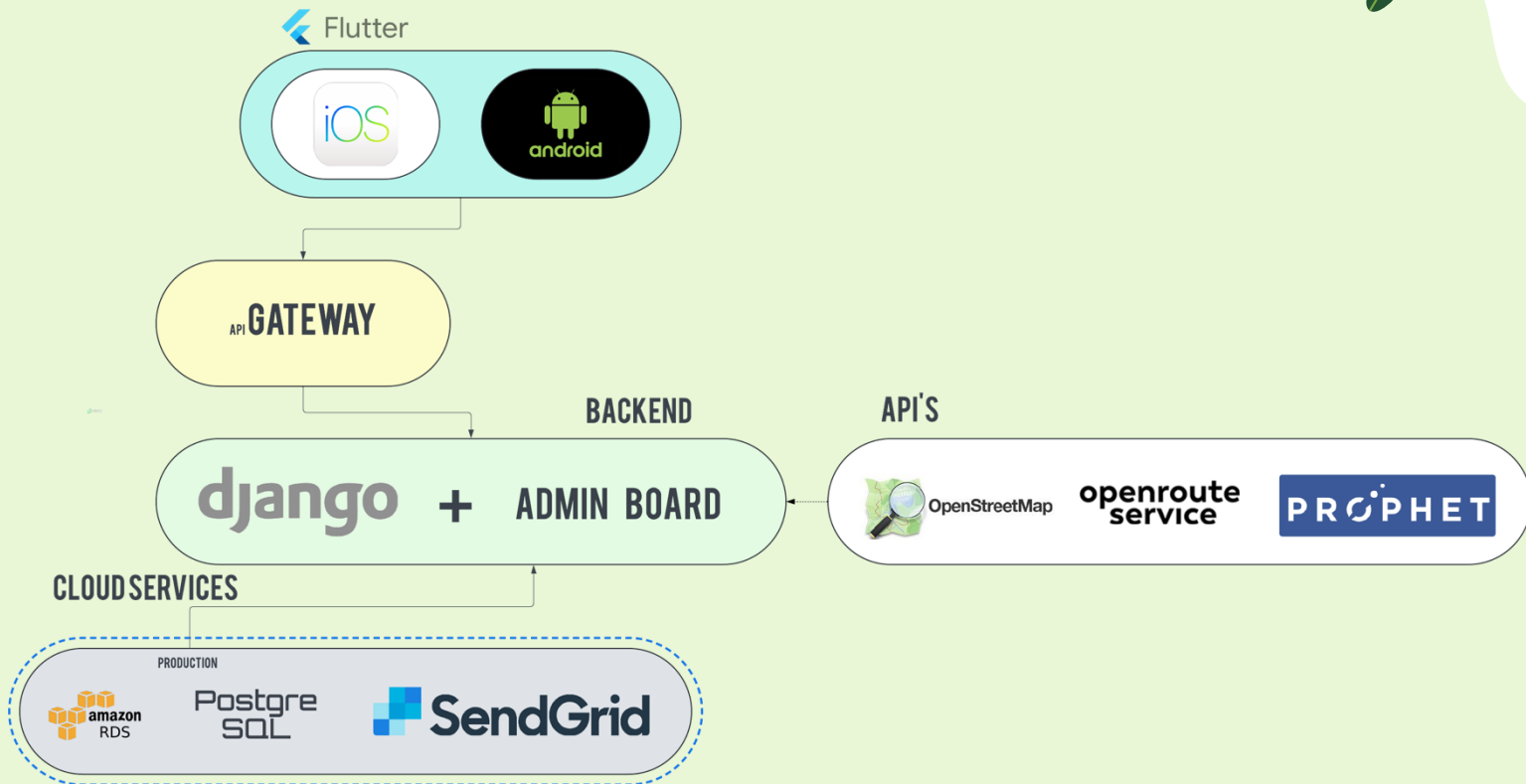
Households/Businesses



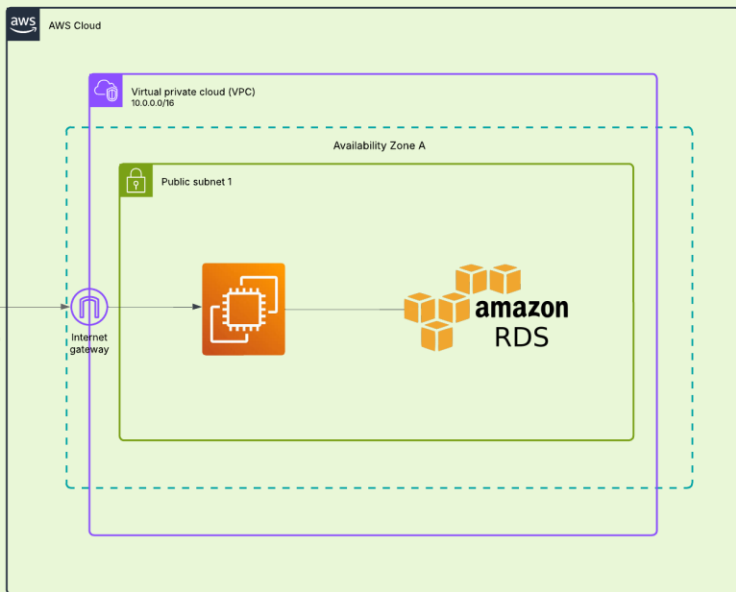
As a **household/business**, I want an **appropriate frequency of waste collection** so that the **bins are not overflowing**.



ARCHITECTURE



CLOUD ARCHITECTURE



DOMAIN AND DNS CONFIGURATION

- Use amazon's route 53 DNS service to direct traffic to our site (devnest.software)
- Use A records to map our domain name to an Elastic IP associated with our EC2 instance.

HTTP & HTTPS TRAFFIC ROUTING

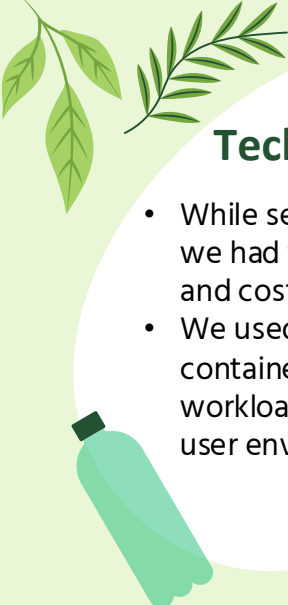
- NGINX reverse proxy handles all HTTP and HTTPS traffic (enforced) and reroutes them to our backend server running inside Docker.
- Security certificates are managed using AWS certificate Manager (ACM)

AMAZON RDS – POSTGRESQL

- We use Amazon RDS for data storage running in a single Availability Zone (cost efficient)
- Use automated backups for disaster recovery

AUTOMATED DEPLOYMENT WITH CRON JOB

- We have a CRON job scheduled daily running on our EC2 instance
- This CRON job runs a new container with the updated image pulled from the Amazon Elastic Container Registry (ECR)

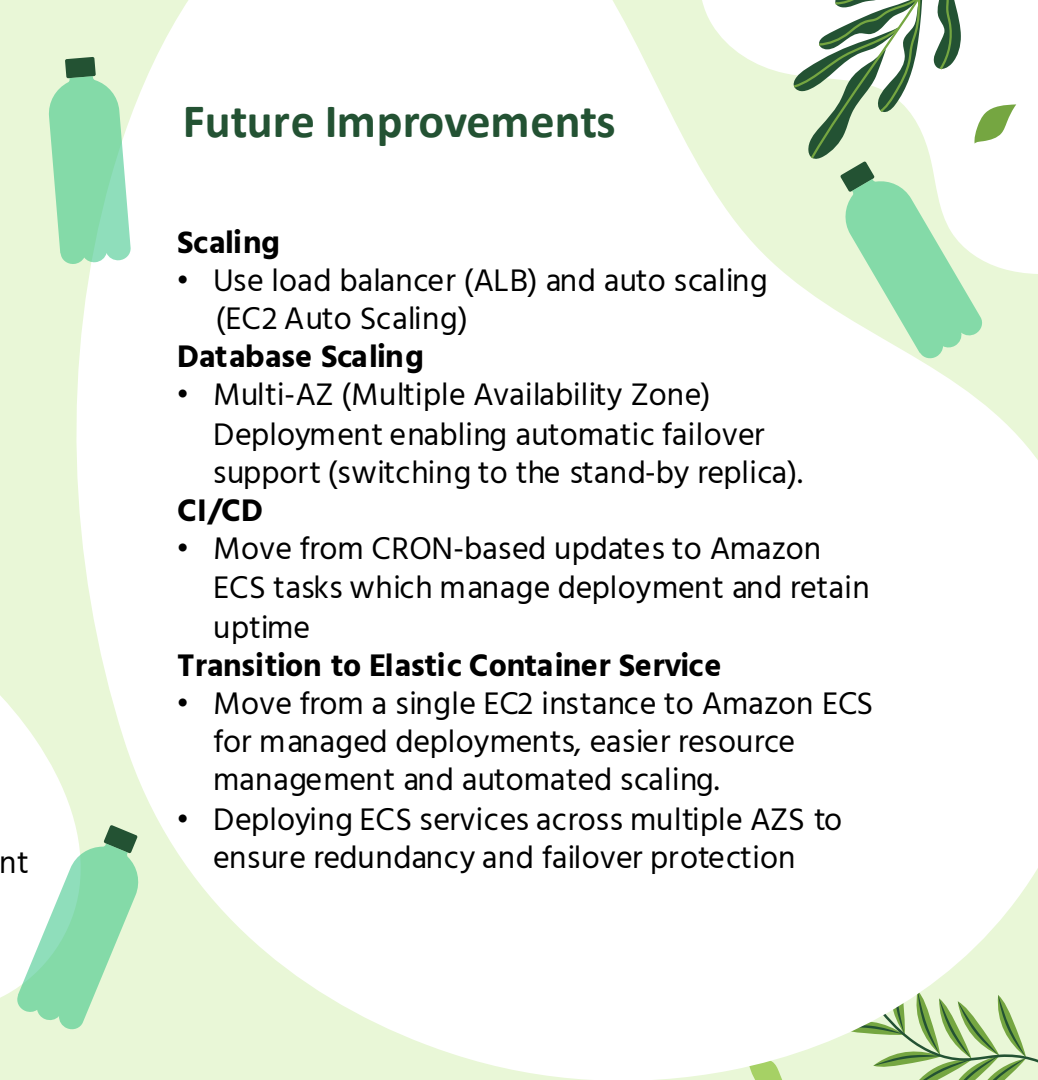


Technical Considerations

- While setting up the backend infrastructure we had to compromise between performance and cost
- We used EC2 instead of Fargate (serverless container) to reduce the cost of persistent workload as this more aligns with production user environment.

FINAL RELEASE

- For our final release we want our infrastructure to be fully scalable, fault tolerant and production-ready
 - Use ECS for fully automated deployment with rolling updates
- 



Future Improvements

Scaling

- Use load balancer (ALB) and auto scaling (EC2 Auto Scaling)

Database Scaling

- Multi-AZ (Multiple Availability Zone) Deployment enabling automatic failover support (switching to the stand-by replica).

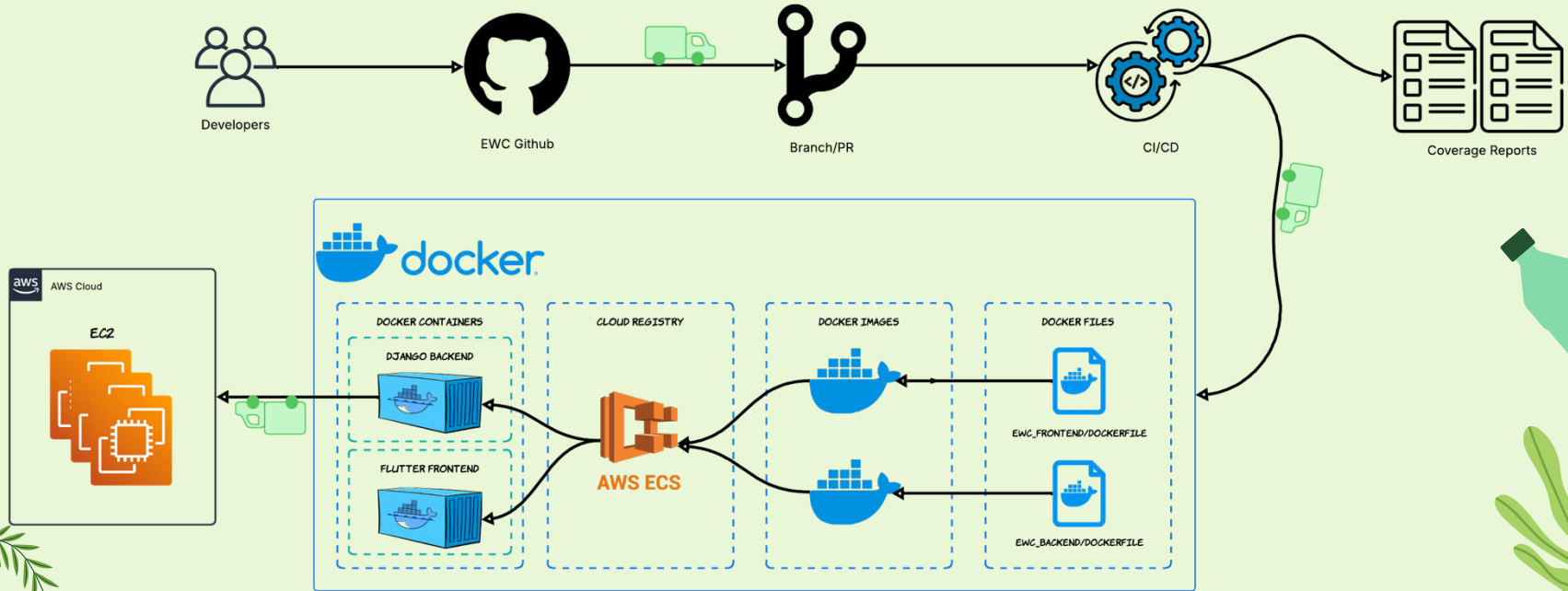
CI/CD

- Move from CRON-based updates to Amazon ECS tasks which manage deployment and retain uptime

Transition to Elastic Container Service

- Move from a single EC2 instance to Amazon ECS for managed deployments, easier resource management and automated scaling.
- Deploying ECS services across multiple AZS to ensure redundancy and failover protection

CONTINUOUS INTEGRATION AND DEPLOYMENT



CI – Coverage

FLUTTER (FRONTEND) COVERAGE

LCOV - code coverage report

Current view: [top level](#)

Test: lcov.info

Test Date: 2025-02-24 10:43:26

Directory	Line Coverage ↕			Function Coverage ↕		
	Rate	Total	Hit	Rate	Total	Hit
lib/lib		4.2 %	24	1	-	
lib/models/lib/models		0.0 %	1	-		
lib/notifiers/lib/notifiers		56.0 %	25	14	-	
lib/screens/login/lib/screens/login		53.4 %	103	55	-	
lib/screens/map/lib/screens/map		44.6 %	148	66	-	
lib/screens/metrics/lib/screens/metrics		88.9 %	198	176	-	
lib/screens/register/lib/screens/register		57.4 %	54	31	-	
lib/screens/route-schedule/lib/screens/route-schedule		0.0 %	54	-		
lib/screens/settings/lib/screens/settings		39.1 %	64	25	-	
lib/screens/splash/lib/screens/splash		91.7 %	36	33	-	
lib/services/lib/services		6.2 %	227	14	-	
lib/theme/lib/theme		0.0 %	5	-		
lib/widgets/graphs/bar-graph/lib/widgets/graphs/bar-graph		98.5 %	66	65	-	
lib/widgets/graphs/line-graph/lib/widgets/graphs/line-graph		83.3 %	48	40	-	
lib/widgets/lib/widgets		61.9 %	247	153	-	

Generated by: [LCOV version 2.0-1](#)

DJANGO (BACKEND) COVERAGE

Coverage report: 82%

[Files](#) [Functions](#) [Classes](#)

coverage.py v7.6.11, created at 2025-02-24 23:54 +0000

File	statements	missing	excluded	coverage
ewc_admin/__init__.py	0	0	0	100%
ewc_admin/settings.py	34	2	0	94%
ewc_admin/urls.py	16	0	0	100%
ewc_core/__init__.py	0	0	0	100%
ewc_core/admin.py	7	0	0	100%
ewc_core/apps.py	4	0	0	100%
ewc_core/forms.py	7	0	0	100%
ewc_core/migrations/__init__.py	0	0	0	100%
ewc_core/migrations/0001_initial.py	7	0	0	100%
ewc_core/migrations/0002_alter_routeenvdata_date.py	4	0	0	100%
ewc_core/migrations/0002_alter_stopcollection_date.py	4	0	0	100%
ewc_core/migrations/0003_alter_routeenvdata_date.py	4	0	0	100%
ewc_core/migrations/0004_merge_20250224_1115.py	4	0	0	100%
ewc_core/ml_model/data.py	15	9	0	40%
ewc_core/ml_model/model.py	45	36	0	20%
ewc_core/models.py	43	3	0	93%
ewc_core/serializers.py	41	0	0	100%
ewc_core/tests/__init__.py	0	0	0	100%
ewc_core/tests/test_api.py	71	0	0	100%
ewc_core/tests/test_models.py	11	0	0	100%
ewc_core/views.py	85	24	0	72%
manage.py	11	2	0	82%
Total	413	76	0	82%

coverage.py v7.6.11, created at 2025-02-24 23:54 +0000

SEP TESTING DAY



USER FEEDBACK

7/26 users tried clicking the location markers on the map for more information on the stops.

4/26 users tried clicking on the stops on the schedule page.

Many people were confused or wanted further clarification on what was being displayed on the schedule page.

Users found a legacy version of the reset password page that can be found on the settings page.

Many people wanted distances to be automatically calculated.

It was suggested that we have more instructions to guide navigation.

Many people liked the option of dark mode.

Some people thought that the map overlay was hard to read and not clear.

Some people suggested that the app should remember the MPG, so you don't have to enter it every time.

Some people wanted there to be a dialogue box that appears after you successfully register.

People thought the bar graph was the clearest graph.

People thought that app overall was aesthetically pleasing.

Most people thought the app had a low frustration level.

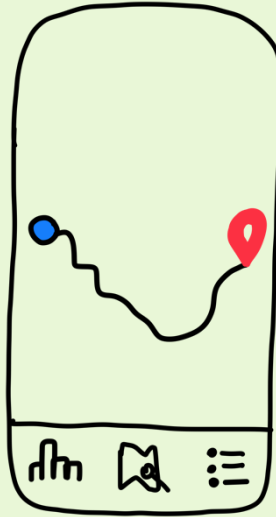
Most people thought the app had a low mental load.



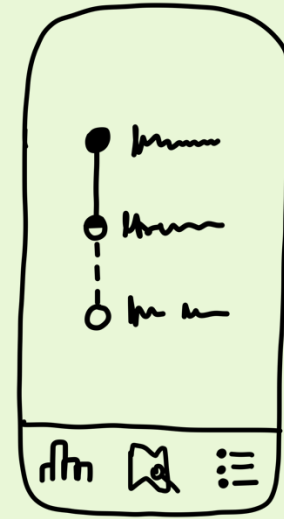
App Design – where we started



Metrics Page



Map Page

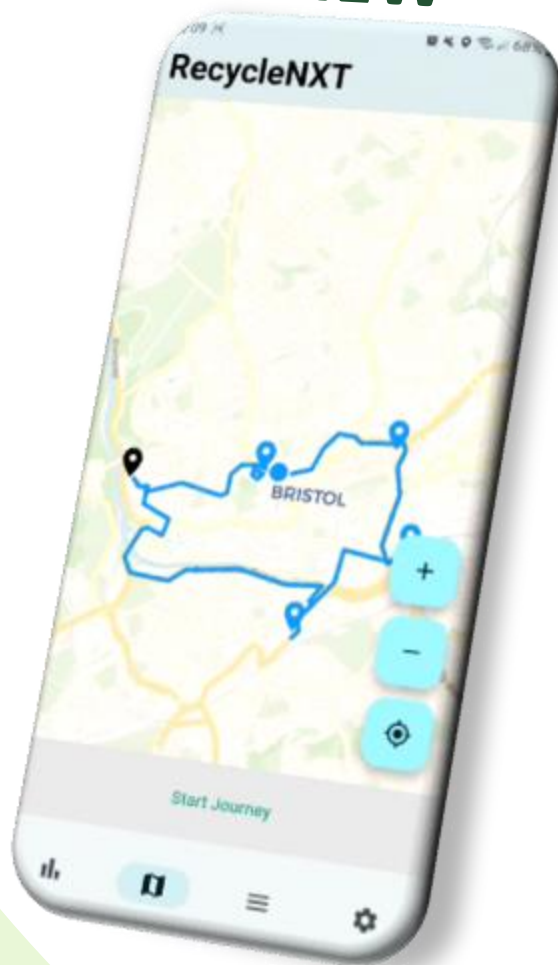


Schedule Page

Old



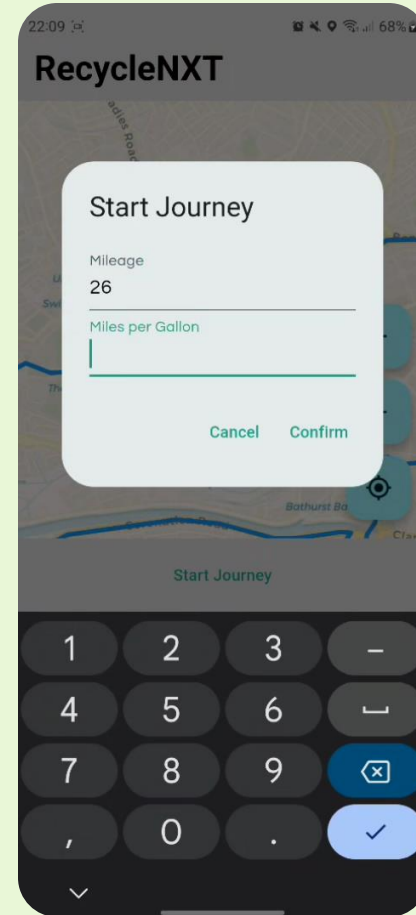
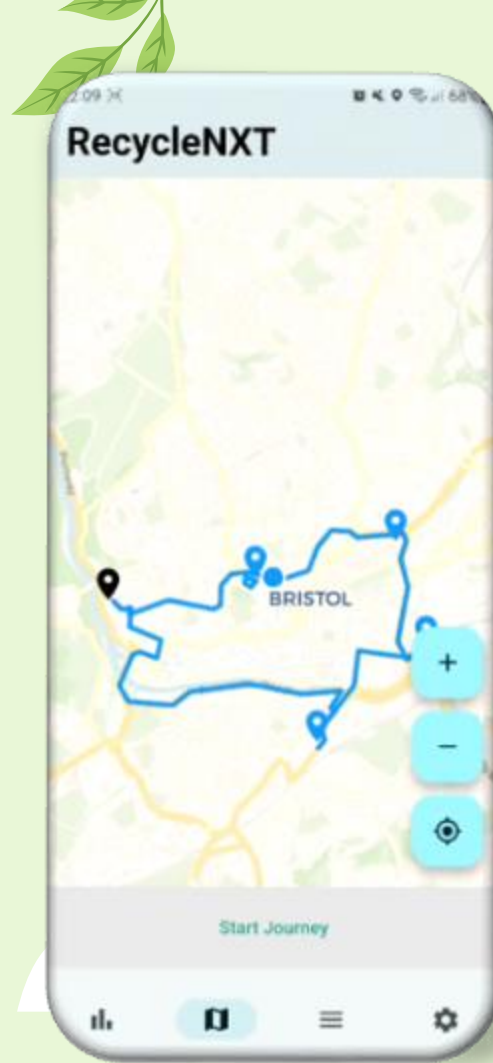
NEW



Design

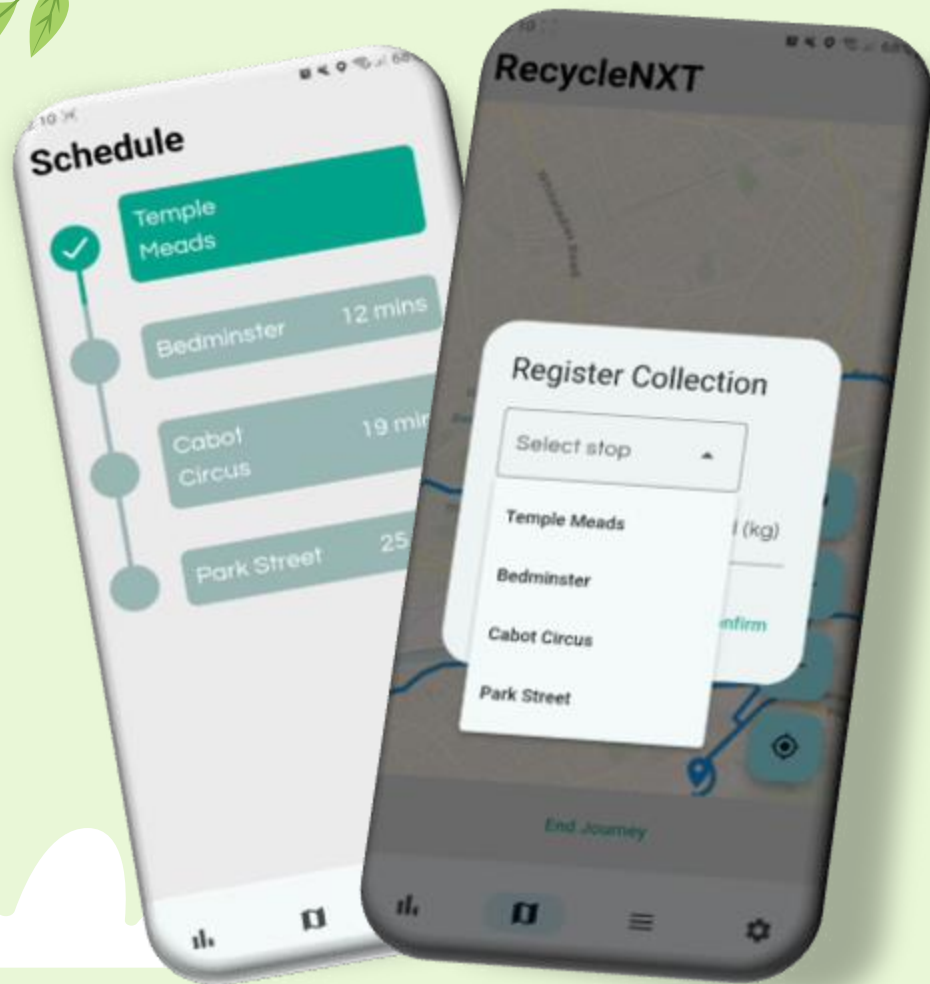
New Features

- Multi-stop Optimised route plotting
- Live location tracking
- Map interactions



Schedule and Map Page Integration

Mark completed stops

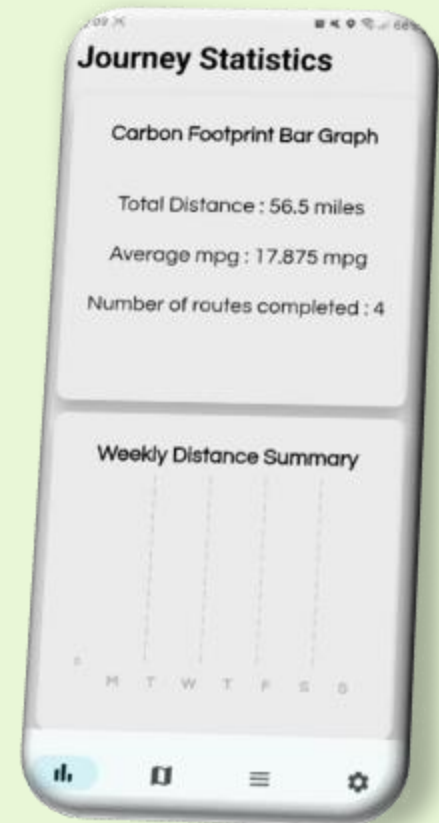
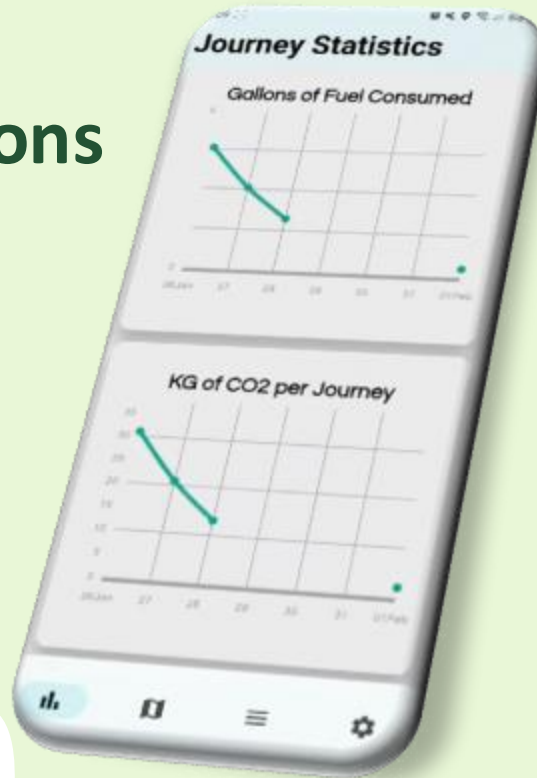


New graphs, data insights and predictions

New ways of tracking data

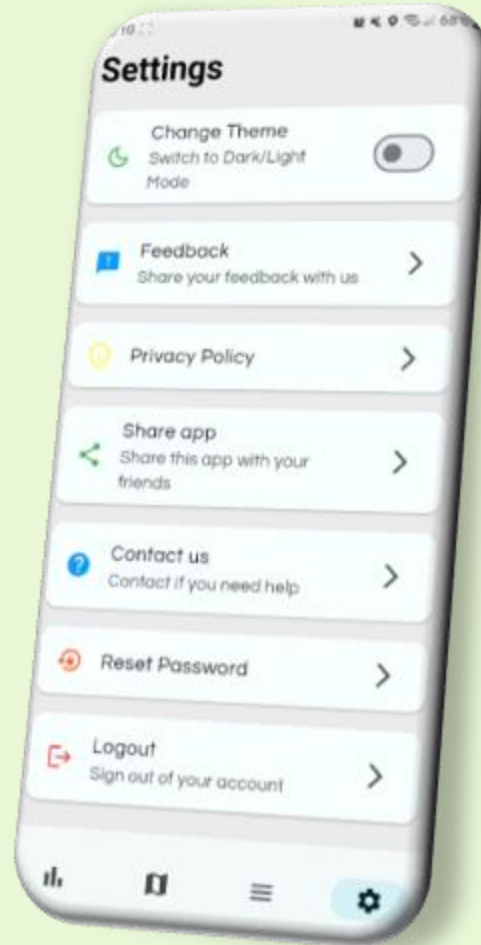
More graphs

PROPHET



New Settings Page

For managing app
preferences and
settings



Admin Page – stops management

Waste Collection Management

Efficiently manage and optimize waste collection routes, track stops, and help the planet breathe.

Stops List

[+ Add New Stop](#)

- Temple Meads**
Lat: 51.4492, Lng: -2.581 · Max Weight: 10
[Edit](#) [Delete](#)
- Bedminster**
Lat: 51.4409, Lng: -2.5963 · Max Weight: 9
[Edit](#) [Delete](#)
- Cabot Circus**
Lat: 51.4585, Lng: -2.5853 · Max Weight: 20
[Edit](#) [Delete](#)
- Park Street**
Lat: 51.4553, Lng: -2.605 · Max Weight: 8
[Edit](#) [Delete](#)

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Create New Stop

Location name

Latitude

Longitude

Max weight

Next collection due date ⓘ This date is usually set automatically by the model's prediction. Only change this if necessary for unusual scenarios.

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DEMO

CHALLENGES



- **Challenge:** Coordinating tasks between group members slowed development

- **Solution:** Increased discussion on pull requests to better align and develop ideas



- **Challenge:** Varying commit sizes lead to unequal commit graphs.

- **Solution:** Focused on making commit numbers accurately represent our equal contribution and division of work



- **Challenge:** Technical issues on testing day meant we did not receive full feedback on parts of the app

- **Solution:** Will conduct more testing to get feedback on these parts



- **Challenge:** Vague feedback from client after MVP

- **Solution:** Group meeting after Christmas to plan out our goals for the app



OUR PROCESS



WEEKLY MEETINGS

Wednesday and Thursday meetings to check progress and decide what to do next.

DISCORD

Using a discord chat with channels to organise our conversations.



AGILE WORKFLOW

Separating jobs into manageable tasks and organising them into sprints

GITHUB



Implementing templates early on to ensure detailed issues and pull requests.



MID PROJECT REVIEW

After Christmas we met to discuss the state of our project and reevaluated our process.





THANK YOU!

